FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Advanced)-2020

PAPER – 1

TEST DATE: 17-11-2019

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B.**

1. ***Section-A (01– 7, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*
   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section-A (08 – 13, 26 – 31, 44 – 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S). Each multiple choice question with 4 options has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section–B (14 – 18, 32 – 36, 50 – 54)*** *contains 15 Numerical based questions with answers as numerical value from 0 to 9 and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

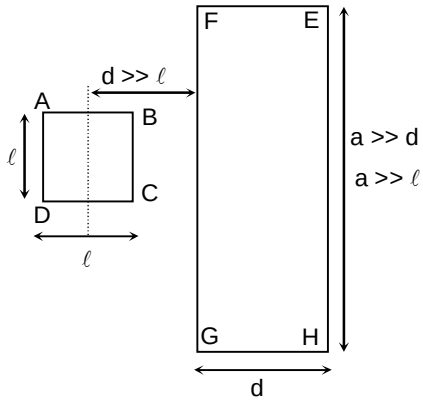
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A square conducting loop ABCD and a rectangular conducting loop EFGH are placed on the same plane as shown in the figure. The loop ABCD is placed symmetrically with respect to EFGH such that its central line is at a distance d from the loop EFGH. Side length 'ℓ' of small loop ABCD is much smaller than the length 'a' and width 'd' of the bigger loop EFGH. A source is applied on loop ABCD, such that it sets up a current I = kt in it. Here k is a positive constant and t is time. Then choose the correct option(s).



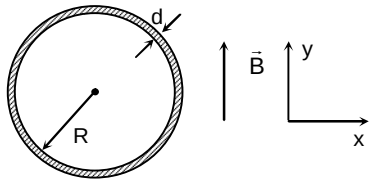
(A) If the current in the loop ABCD is in clockwise sense then the induced current in the loop EFGH due to this current is in anticlockwise sense.

(B) If the current in the loop ABCD is in clockwise sense then the induced current in the loop EFGH due to this current is in clockwise sense.

(C) emf induced in the loop EFGH due to current in the loop ABCD is $\frac{\mu_0 \ell^2 k}{4\pi d}$.

(D) emf induced in the loop EFGH due to current in the loop ABCD is $\frac{3\mu_0 \ell^2 k}{4\pi d}$.

2. A thin walled non-magnetic spherical shell with wall thickness d and radius R (R >> d) is placed in a uniform magnetic field $\vec{B} = B_0 \hat{j}$, which slowly changes as $\frac{dB_0}{dt} = \alpha$, where α is a positive constant. Electrical resistivity of the material of the shell is ρ.



Then choose the correct option(s).

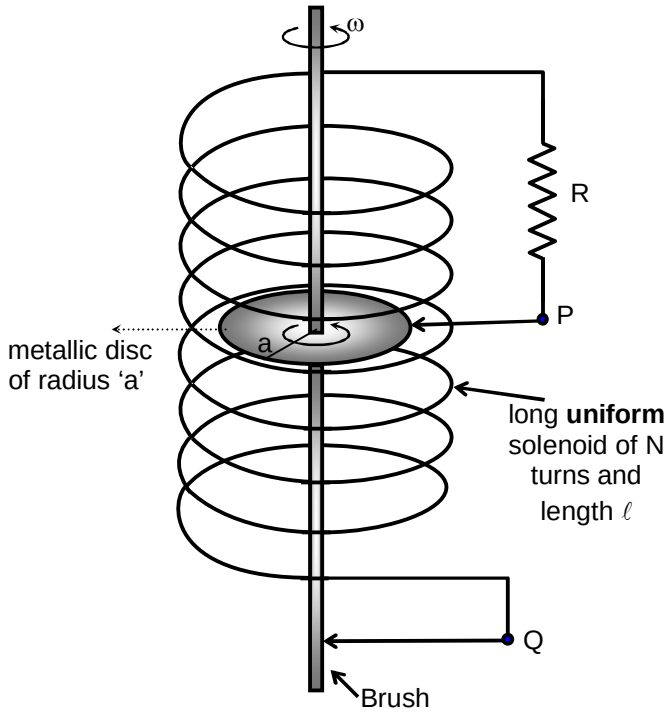
(A) Induced magnetic moment of the shell is towards $+\hat{j}$ direction

(B) Induced magnetic moment of the shell is towards $-\hat{j}$ direction

(C) Induced magnetic moment of the shell is $\frac{3\pi}{2\rho} dR^4 \alpha$

(D) Induced magnetic moment of the shell is $\frac{2\pi}{3\rho} dR^4 \alpha$

3. A metallic disc of radius 'a' mounted on a rigid axle is rotating with a constant angular velocity ω inside a long solenoid of inductance L whose two ends are connected to the rotating disc by two brush contacts as shown. The total resistance of the whole circuit is R. Then



(A) The induced emf between terminals P and Q is $\frac{\mu_0 Ni\omega a^2}{2\ell}$, when current in the coil is i.

(B) Current in the coil can increase or decrease with time depending upon the value of ω.

(C) ω_{min} for increasing the current in the coil always is $\frac{\ell R}{2\mu_0 Na^2}$

(D) ω_{min} for increasing the current in the coil always is $\frac{2\ell R}{\mu_0 Na^2}$

4. A solenoid of radius r is made of thin metallic wire rings with resistance of each ring R. The rings have been put in a uniform way on a very long glass cylinder which has vacuum inside it. The solenoid and glass cylinder are glued to each other. The number of rings per unit length is n and the planes containing the rings are perpendicular to the symmetry axis of the cylinder. At a moment the cylinder starts rotating around its symmetry axis with an angular acceleration α. The magnetic field B at the centre of the cylinder (after a long time) is ($\frac{e}{m}$ is specific charge of electron).

(A) $\frac{\mu_0 nmr^2\alpha}{eR}$

(B) $\frac{\pi\mu_0 nmr^2\alpha}{eR}$

(C) $\frac{2\pi\mu_0 nmr^2\alpha}{eR}$

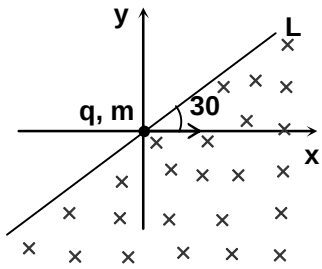
(D) $\frac{4\pi\mu_0 nmr^2\alpha}{eR}$

5. During the melting of a slab of ice at 273 K at the atmospheric pressure.
   (A) Positive work is done by the ice-water system on the atmosphere.
   (B) Positive work is done on the ice-water system by the atmosphere.
   (C) The internal energy of the ice-water system increases.
   (D) The internal energy of the ice-water system decreases.

6. An electric dipole having dipole moment $\vec{P} = \left(3\hat{i} + 4\hat{j}\right) \times 10^{-9}$ Cm is placed in a uniform electric field $\vec{E} = \left(-60\hat{i} - 80\hat{j}\right)$ N/C.
   (A) Electric field applies zero net force on the dipole
   (B) Electric field energy of the dipole in the electric field is -5×10^{-7} J.
   (C) The dipole is in stable equilibrium in the electric field.
   (D) The dipole is in unstable equilibrium in the electric field.

7. A charged particle of charge q and mass m is entering at origin with a velocity $v\hat{i}$ in a uniform magnetic field $-B\hat{k}$ which is extended upto infinite at right of the line L. Select the correct option(s)

   Given that $\frac{mv}{qB} = 1$ meter. and v = 1 m/s.



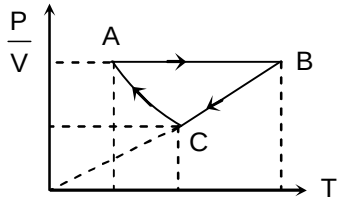
   (A) Velocity of the particle makes an angle 60° with x-axis while it is coming out of the magnetic field.
   (B) The co-ordinates of point of exit is $\left(\frac{1}{2}m, \frac{\sqrt{3}}{2}m\right)$
   (C) The co-ordinates of point of exit is $\left(\frac{\sqrt{3}}{2}m, \frac{1}{2}m\right)$.
   (D) Time spent by the particle in the magnetic field is $\frac{\pi}{3}$ sec.

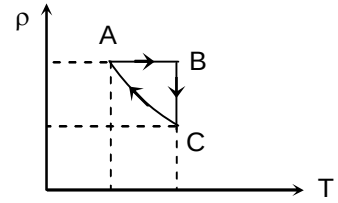
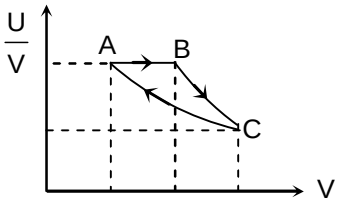
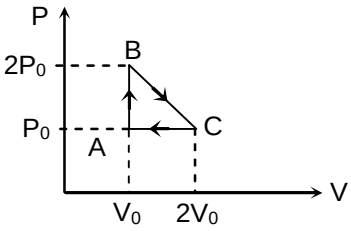
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer questions 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

Here P, V, T ρ, U have usual meaning of pressure, volume, temperature, density and Internal energy of gas.

Column – 1		Column – 2		Column – 3	
(I)	 CA – Rectangular Hyperbola	(i)	$W_{AB} > 0$	(P)	$\Delta U_{AB} > 0$

(II)	 CA – Rectangular Hyperbola	(ii)	$W_{AB} < 0$	(Q)	$\Delta U_{BC} = 0$
(III)	 CA – Rectangular Hyperbola	(iii)	$W_{BC} > 0$	(R)	$\Delta U_{AB} < 0$
(IV)		(iv)	$W_{CA} < 0$	(S)	$\Delta U_{CA} < 0$

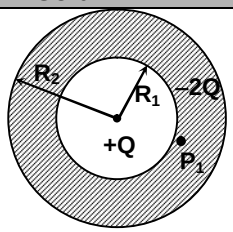
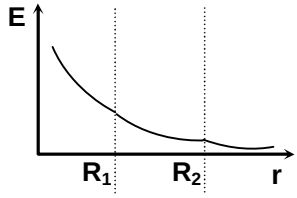
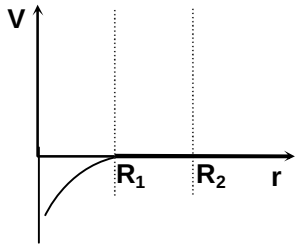
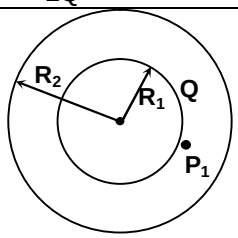
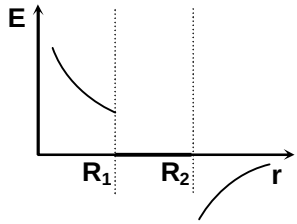
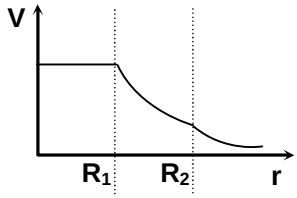
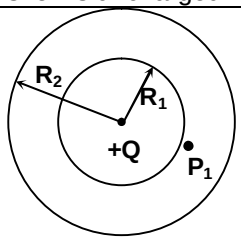
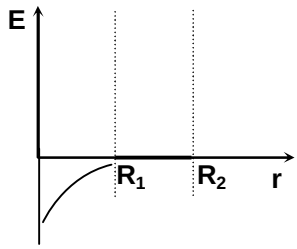
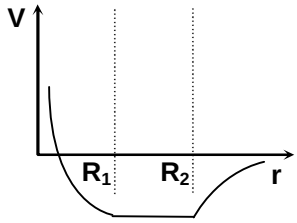
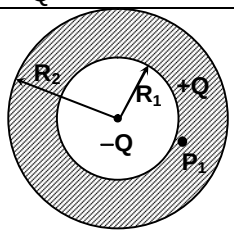
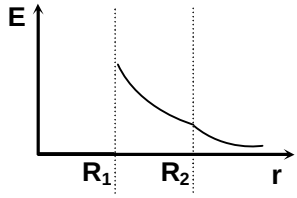
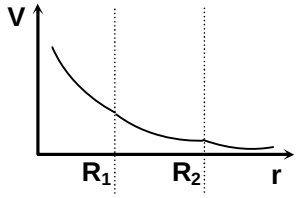
8. In a cyclic process W_{AB} and ΔU_{AB} are both non zero values and have same sign.
   (A) (I) (ii) (R)
   (B) (II) (i) (P)
   (C) (III) (i) (P)
   (D) (IV) (ii) (R)

9. In a cyclic process ΔQ_{BC} must be positive.
   (A) (I) (iii) (Q)
   (B) (II) (iii) (Q)
   (C) (III) (iii) (Q)
   (D) (IV) (iii) (Q)

10. In a cyclic process, $\Delta W_{\text{any path}} + \Delta U_{\text{any path}} < 0$ (any path means any of AB or BC or CA)
   (A) (I) (ii) (R)
   (B) (II) (iv) (S)
   (C) (III) (iv) (S)
   (D) (IV) (i) (R)

Answer questions 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.

Column –1 is having charged systems, Column –2 is having variation of electric field for any charged system of column –1 and Column –3 is having variation of electric potential (taking potential at infinity to be zero) for any charged system of column –1

Column – 1		Column – 2		Column – 3	
(I)	 Point charge at center = Q and charge on thick conducting shell = –2Q	(i)		(P)	
(II)	 Two thin conducting shell, out of which inner shell is having charge Q and outer shell is uncharged	(ii)		(Q)	
(III)	 Two thin uncharged conducting shell and Point charge at centre = Q	(iii)		(R)	
(IV)	 Point charge at center = –Q, Charge on thick conducting shell = +Q	(iv)		(S)	

In each charged system Q = 1 μC, R_1 = 10 cm, R_2 = 20 cm. Each of the shells or spheres are conducting. Variation of electric field and potential is given as a function of radial distance r. point P_1 in each figure is at distance $r = R_1 + \delta$ (where $\delta \to 0$)

11. Electric field at point P_1 is $900\frac{KV}{m}$ for a charged system. Choose the correct option for that charged system.
(A) (III) (i) (Q)
(B) (III) (iv) (S)
(C) (II) (iv) (Q)
(D) (II) (i) (P)

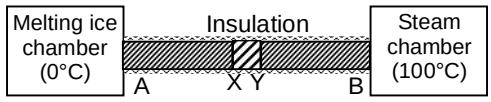
12. Electric potential of a conducting shell (thin or thick) is -45 KV for a charged system. Choose the correct option for that charged system.
(A) (I) (ii) (R)
(B) (I) (ii) (P)
(C) (III) (i) (R)
(D) (III) (i) (P)

13. Electrical potential energy stored in outside space (from $r = R_2$ to $(r \to \infty)$ is 2.25×10^{-2}J
(A) (I) (ii) (Q)
(B) (II) (iv) (S)
(C) (III) (i) (P)
(D) (III) (i) (S)

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

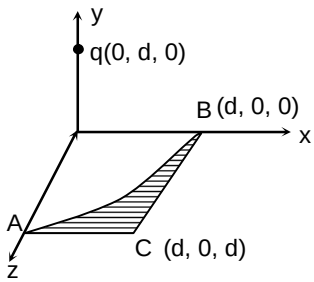
14. A conducting rod is initially at 0°C temperature. Now one end A is kept at melting ice chamber (0°C) and the other end B is kept at steam chamber (100°C). The surface of the rod is well insulated from the outer environment so that heat is only flowing along the length of the rod as shown is the figure.



The cross-sectional area of the rod is 1 cm^2 and length of the rod is 1 m. Thermal conductivity of the rod is 500 W/m-K. Density of the rod is 8000 kg/m^3. Specific heat capacity of the rod is 500 J/kg-K. A small cylindrical element xy (of length 1 cm and cross-sectional area same as that of rod) is examined. It is observed that temperature gradient of the end X is 5°C/m and average temperature of xy part of the rod is rising at 0.05°C/sec. What is the temperature gradient (in °C/m) of end Y of the element?

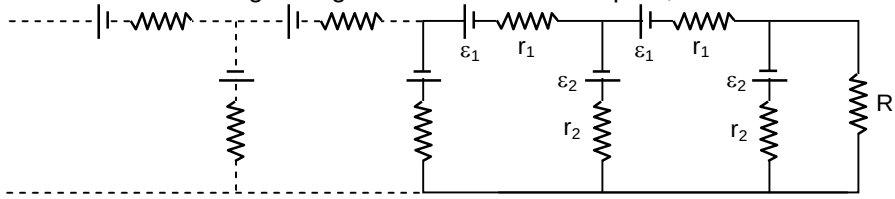
15. A point charge $q = 8.85 \times 10^{-9}$ C is placed at (0, d, 0) where d = 1m. An insulating surface ABCA is on the x-z plane, which is bounded by a quarter circle AB and two equal length straight line BC and CA as shown in the figure. Electric flux through the surface ABCA in Nm^2/C is (nearest integer)

Take $\varepsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^2}$

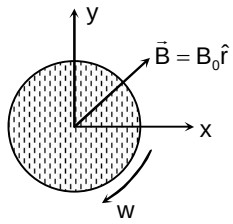


16. Chandrayan-1, the lunar mission launched by India in 2008, had all gold wiring. A particular instrument on board the mission, was operating at the same temperature as the space surrounding the Chandrayan at 3 K. Instrument specifications demanded that no wire in the instrument should offer a resistance more than 7 mΩ. All the instruments on Chandrayan were fabricated in ISRO lab at room temperature of 30°C. If the radius of all wires in the instrument was 0.1 mm (as measured in ISRO lab). If the maximum allowable length (as measured in ISRO lab) of any wire is approximately ℓ meter then find $\frac{1}{\ell}$. $\left(\text{Take } \pi = \frac{22}{7}\right)$. (Assume that resistivity and length of gold wire change linearly with temperature)
You can use these properties of gold.
- Resistivity at room temperature $\rho = 2.2 \times 10^{-8}\ \Omega m$
- Temperature coefficient of resistivity $\beta = 0.0032\ K^{-1}$
- Linear thermal expansion coefficient $\alpha = 1.5 \times 10^{-5}\ K^{-1}$.

17. The network shown consists of an infinite number of cells of emf ε_1 and ε_2 with corresponding internal resistances r_1 and r_2 respectively A resistor R = 1Ω is connected to the end of the network. If the values of the emfs and internal resistances are ε_1 = 1.0 V, ε_2 = 2.0 V, r_1 = 4Ω and r_2 = 6Ω. If the current flowing through the resistor R is I ampere, then find 10 I.



18. A thin non conducting disc of mass M = 2 kg, charge Q = 2×10^{-2}C and radius $R = \frac{1}{6}$m is placed on a frictionless horizontal plane with its centre at the origin of the coordinate system. A non uniform, radial magnetic field $\vec{B} = B_0\hat{r}$ exists in space, where B_0 = 10 T and the disc is rotated with an angular velocity $\omega = x \times 10^2$ rad/sec about an axis passing through its centre and perpendicular to its plane, as shown in the figure. At what value of x the disc will lift off from the surface.

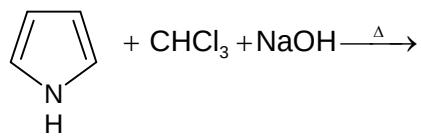


Chemistry

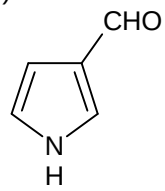
PART – II

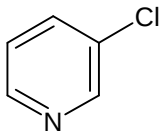
SECTION – A
(More Than One Correct Type)

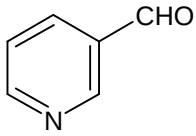
This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

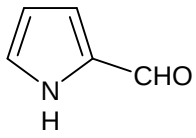
19. 

Product (s) formed is/are

(A) 

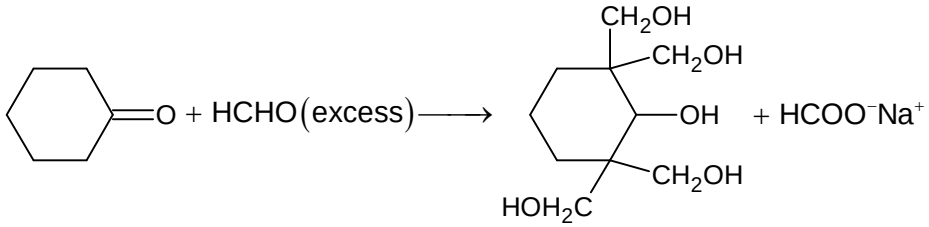
(B) 

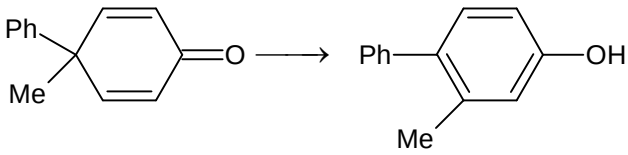
(C) 

(D) 

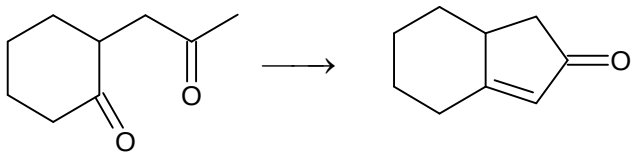
20. Which of the following statement is/are correct?
    (A) Cellulose nitrate is a natural polymer.
    (B) Vulcanization increases rigidity by restricting the movement of the polymer chains.
    (C) HDPE is a linear polymer whereas LDPE is a branched polymer.
    (D) Cellulose is an example of an unbranched polysaccharide.

21. Which of the following transformation(s) is/are possible by aq. NaOH in presence or absence of heating

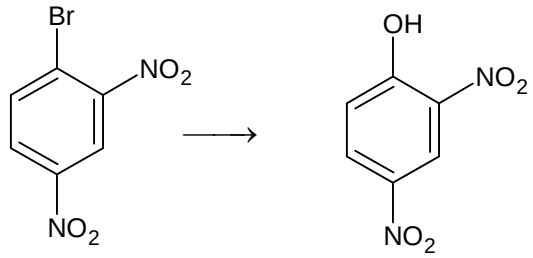
(A) 

(B) 

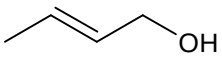
(C)



(D)



22.



Which of the following is/are correct about the above drawn compound?

(A) It gives positive Br_2 – water test.
(B) It gives positive Na-metal test.
(C) It gives negative iodoform test.
(D) It does not give turbidity with Lucas reagent at room temperature.

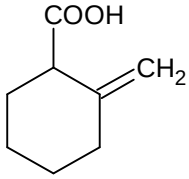
23. $H_2C = CH_2 \xrightarrow{SCl_2} P$

Which of the following statement(s) is/are correct regarding 'P'?

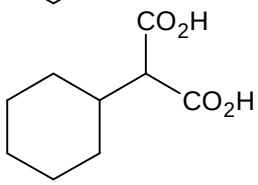
(A) P on treatment with sodium ethoxide produces divinyl sulphide.
(B) It is a chemical warfare agent.
(C) Rate of hydrolysis of P is much more than its corresponding ether analogue.
(D) On complete hydrolysis it produces HCl and diol.

24. Which of the following acids undergo decarboxylation just on heating?

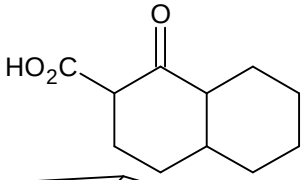
(A)



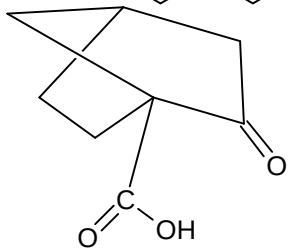
(B)



(C)



(D)



25. How many of the following statements is/are correct?
(A) α-D-(+)-glucopyranose and β-D-(+)-glucopyranose are diastereomers
(B) D-Glucose and D-Mannose are diastereomers
(C) Maleic acid and Fumaric acid are diastereomers
(D) Dextro' form and 'Meso' form of tartaric acid (2,3-dihydroxybutanedioic acid) are diastereromers.

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer Q. 26, Q. 27 and Q. 28 by appropriately matching the information given in the three columns of the following table.

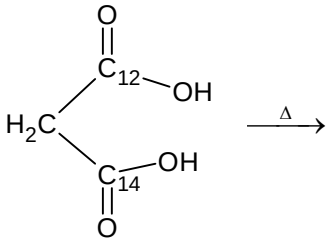
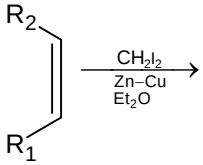
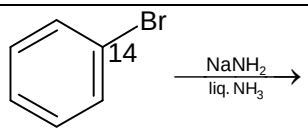
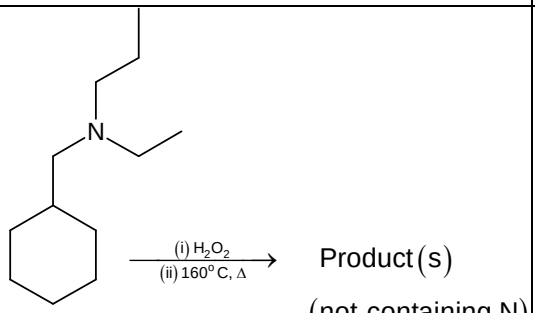
Column 1		Column 2		Column 3	
(I)	Orlon	(i)	High impact resistance and mechanical toughness	(P)	Elastometer
(II)	Buna-N	(ii)	Have good resistance to stains, chemicals	(Q)	Homopolymer
(III)	Dacron	(iii)	Resistant to the action of petrol, lubricating oil	(R)	Terpolymer
(IV)	ABS	(iv)	Crease resistant	(S)	Thermoplastic or thermoset depend on the chemical structure

26. Which of the following is only correct combination?
(A) (I) (iv) (S)
(B) (II) (iii) (P)
(C) (IV) (i) (P)
(D) (I) (iv) (S)

27. Which of the following is only correct combination?
(A) (IV) (i) (R)
(B) (II) (ii) (Q)
(C) (III) (iv) (Q)
(D) (I) (ii) (P)

28. Which of the following is only correct combination?
(A) (II) (ii) (S)
(B) (III) (ii) (R)
(C) (IV) (i) (P)
(D) (I) (ii) (Q)

Answer Q. 29, Q. 30 and Q. 31 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)		(i)	E_{1CB} involved during the reaction	(P)	One product
(II)		(ii)	Proceeds via six membered cyclic TS	(Q)	Three products
(III)		(iii)	Proceeds via five membered cyclic TS	(R)	Four products
(IV)		(iv)	Butterfly type transition state	(S)	Two products

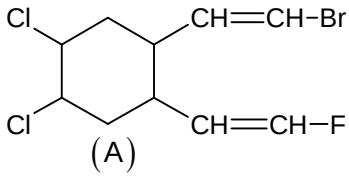
29. Which of the following is only correct combination?
(A) (III) (ii) (R)
(B) (II) (iv) (S)
(C) (IV) (i) (P)
(D) (I) (iii) (S)

30. Which of the following is only correct combination?
(A) (III) (i) (S)
(B) (II) (i) (P)
(C) (IV) (ii) (P)
(D) (I) (iii) (R)

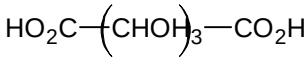
31. Which of the following is only correct combination?
(A) (III) (iv) (S)
(B) (IV) (iii) (Q)
(C) (IV) (ii) (S)
(D) (I) (ii) (P)

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

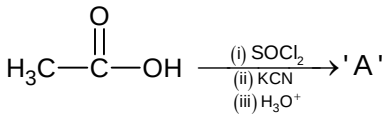
32. 

(A)



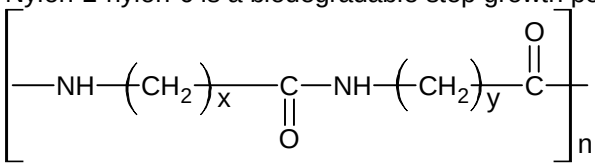
(B)

If number of stereoisomers for (A) is 2^x, and if number of stereoisomers are possible for (B) is 2^y. Then value of x/y would be?

33. 

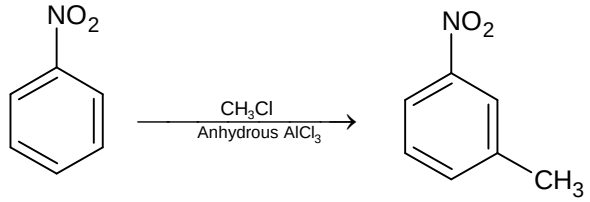
How many of the following statements is/are correct about 'A'?
(1) It forms oxime with hydroxylamine
(2) It can be prepared by the oxidation of propylene glycol by $KMnO_4$.
(3) It is stronger acid than propanoic acid
(4) It forms phenyl hydrazone with phenyl hydrazine
(5) It can be prepared by hydrolysis of α, α-dibrompropanoic acid
(6) When it is heated with dil. H_2SO_4 in a sealed tube, decarboxylation takes place
(7) When it is heated with conc. H_2SO_4, decarbonylation takes place
(8) It gives positive result in the iodoform reaction
(9) It decolourises bromine water and $KMnO_4$ solution.

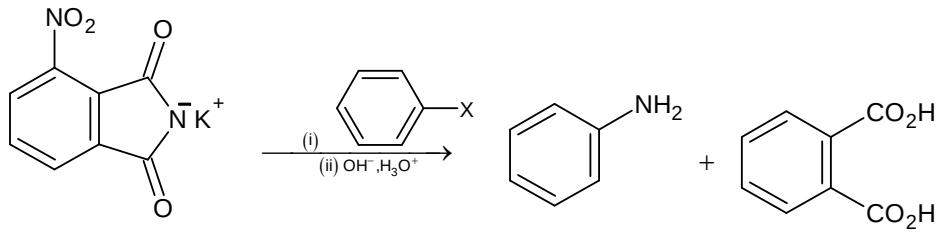
34. Nylon-2-nylon-6 is a biodegradable step growth polymer. It can be expressed as



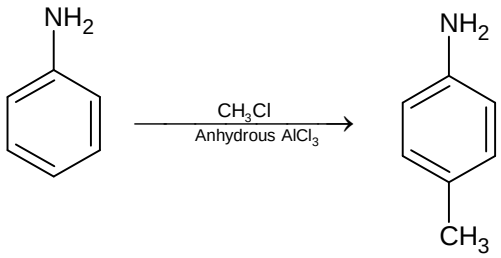
Value of y + x would be?

35. How many of the following reaction is not possible or produce very poor yield?

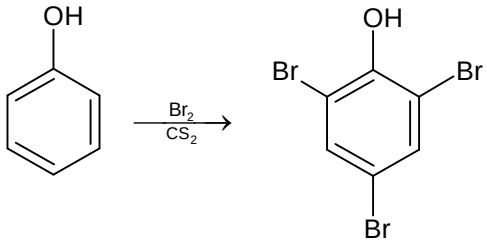
(i) 

(ii) 

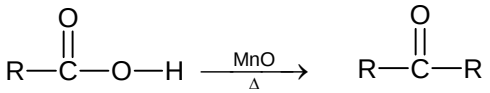
(iii)



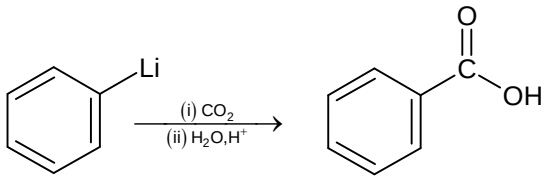
(iv)



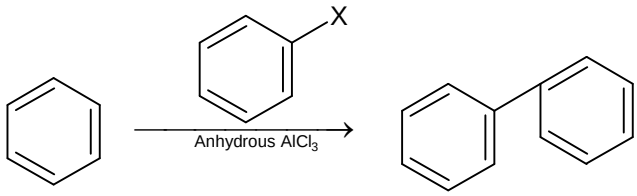
(v)



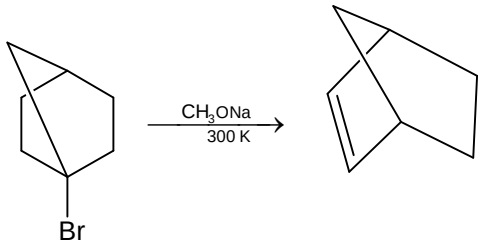
(vi)



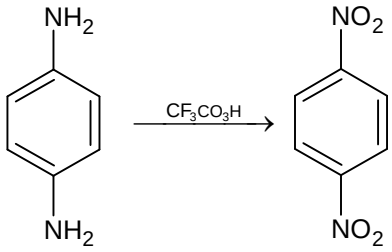
(vii)



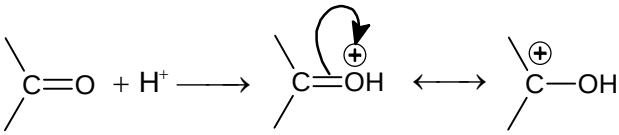
(viii)



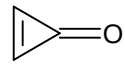
(ix)

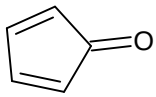


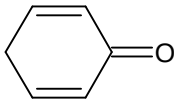
36. Protonation of carbonyl group forms a cation with significant charge on the carbon atom

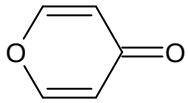


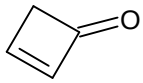
How many of the following carbonyl compounds form an aromatic cation on protonation?

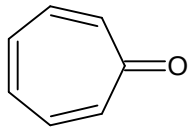
1. 

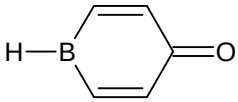
2. 

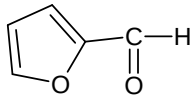
3. 

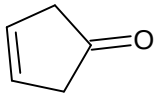
4. 

5. 

6. 

7. 

8. 

9. 

Mathematics

PART – III

SECTION – A
One OR More Than One Choice Type

This section contains **07 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

37. Let the ordered pair of integers (x_i, y_i) where i = 1, 2, ..., n satisfy the equation $\tan^{-1} x + \tan^{-1} 8 = \tan^{-1} y + \frac{\pi}{2}$. For the polygon so formed in xy plane which of the following is/are true?
(A) It is a rectangle
(B) It is not cyclic
(C) It is symmetric about line y = x
(D) It is symmetric about line x + y = 0

38. Let A = {xy = c| c $\in$ R – {0}}. Let $E_C = 0$ represent locus of point P_c such that sum of slopes of normals drawn from it to xy = c is equal to sum of ordinates of feet of normals. Let F be the set of points of intersection of $E_c = 0$ with xy = c (for parameter 'c'), then
(A) (1, 1) is an element of F
(B) (2, 1) is not an element of F
(C) (0, 0) is an element of $E_c \cap F \ \forall$ c $\in$ R – {0}
(D) F is a set of hyperbolas

39. Which of the following is/are correct with respect to angle bisectors of pair of lines $3x^2 – 4y^2 – xy – 12x + 16y = 0$
(A) Angle bisector containing point (2, 1) is the obtuse angle bisector
(B) Equation of acute angle bisector is $(3\sqrt{2}+5)x-(4\sqrt{2}-5)y-20=0$
(C) Equation of obtuse angle bisector is $(3\sqrt{2}+5)x-(4\sqrt{2}-5)y-20=0$
(D) Angle bisectors are perpendicular with respect to each other

40. For the curves $2x^2 + y^2 = 2$ and $y^2 – x^2 = –2$ which of the following is/are true?
(A) Equation of a pair of common tangents is $4x^2 + y^2 – 4xy – 6 = 0$
(B) Equation of a pair of common tangents is $4x^2 + y^2 + 4xy – 6 = 0$
(C) Length of common tangent is $\sqrt{\frac{5}{3}}$
(D) Length of common tangent is $\sqrt{\frac{10}{3}}$

41. Let $E_1 : \frac{x^2}{4}+\frac{y^2}{3}=1$ and $E_2 : \frac{(x-4)^2}{4}+\frac{y^2}{3}=1$. If E_2 rolls on E_1, without slipping, then which of the following is/are true?
(A) Locus of centre of E_2 is a polynomial of fourth degree
(B) Locus of one of the focus of E_2 is polynomial of third degree
(C) Locus of centre of E_2 is a closed curve
(D) Locus of one of the focus of E_2 is a closed curve

42. Let O be circumcenter of ΔABC and AO, BO, CO meet circumcircle of ΔABC in D, E, F respectively. If areas of ΔABC, ΔDBC, ΔEAC, ΔFAB are Δ, Δ_1, Δ_2, Δ_3 and radius of their circumcircle are R, R_1, R_2, R_3 respectively, then (where A, B, C and a, b, c have usual meaning)

(A) $\Delta_1 + \Delta_2 + \Delta_3 = \Delta$

(B) $\Delta_1 + \Delta_2 + \Delta_3 = 2R^2 \sin A \sin B \sin C$

(C) $\dfrac{a}{R_1}+\dfrac{b}{R_2}+\dfrac{c}{R_3}=\dfrac{4\Delta}{R^2}$

(D) $\dfrac{a}{R_1}+\dfrac{b}{R_2}+\dfrac{c}{R_3}=\dfrac{abc}{R^2}$

43. $\prod_{i=1}^{n} P_i^{a_i \cos^2 a_i} = 2019$ where P_i represents the ith prime number

(i.e., for example $P_1 = 2$, $P_2 = 3$, $P_3 = 5$ .....) such that $P_n < 2019$ and $a_i \in R$. If elements of set A be solution of the equation represented by ordered n-tuple (a_1, a_2, a_3 ....., a_n), then

(A) number of elements in set A is finite

(B) if $a_i \le 9 \ \forall$ i = 1, 2, ..., n, then number of element in set A is 25

(C) if $a_i \le 0 \ \forall$ i = 1, 2, ..., n, then A is null set

(D) if $a_i \le 9 \ \forall$ i = 1, 2, ..., n, then A is singleton set

(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer Q.44, Q.45 and Q.46 by appropriately matching the information given in the three columns of the following table.

Let P_1 P_2 P_3 form an equilateral triangle. Let P_{n+3} be mid-point of P_n and P_{n+1} $\forall$ n $\in$ N. Match the ratios of areas of triangle given in Column-1 and ratio of angles of triangle given in Column-2 to their respective values in Column-3

Column–1	Column–2	Column–3
(I) $\dfrac{\Delta P_{2017}\ P_{2018}\ P_{2019}}{\Delta P_{2019}\ P_{2020}\ P_{2021}}$	(i) $\dfrac{\angle P_{2021}\ P_{2022}\ P_{2023}}{\angle P_{2020}\ P_{2019}\ P_{2018}}$	(P) 1
(II) $\dfrac{\Delta P_{2017}\ P_{2018}\ P_{2019}}{\Delta P_{2020}\ P_{2021}\ P_{2022}}$	(ii) $\dfrac{\angle P_{2001}\ P_{2002}\ P_{2003}}{\angle P_{2020}\ P_{2018}\ P_{2021}}$	(Q) 2
(III) $\dfrac{\Delta P_{2017}\ P_{2019}\ P_{2020}}{\Delta P_{2018}\ P_{2019}\ P_{2020}}$	(iii) $\dfrac{\angle P_{2004}\ P_{2005}\ P_{2003}}{\angle P_{2002}\ P_{2003}\ P_{2004}}$	(R) 4
(IV) $\dfrac{\Delta P_{2021}\ P_{2022}\ P_{2023}}{\Delta P_{2021}\ P_{2023}\ P_{2025}}$	(iv) $\dfrac{\angle P_{2017}\ P_{2018}\ P_{2019}}{\angle P_{2018}\ P_{2019}\ P_{2020}}$	(S) 8

44. Which of the following combination is only CORRECT?

(A) (I) (i) (Q)

(B) (II) (iii) (P)

(C) (I) (iii) (R)

(D) (II) (i) (S)

45. Which of the following combination is only CORRECT?
(A) (III) (ii) (P)
(B) (IV) (i) (R)
(C) (III) (i) (P)
(D) (IV) (iv) (R)

46. Which of the following combination is only CORRECT?
(A) (II) (iv) (P)
(B) (IV) (i) (Q)
(C) (II) (ii) (R)
(D) (IV) (iv) (P)

Answer Q.47, Q.48 and Q.49 by appropriately matching the information given in the three columns of the following table.

Let inverse trigonometric functions be defined as follows: $\sin^{-1}x : [-1, 1] \to \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$;

$\cos^{-1}x : [-1, 1] \to [0, \pi]$; $\tan^{-1}x : R \to \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$; $\text{cosec}^{-1}x : (-\infty, -1] \cup [1, \infty) \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$;

$\cot^{-1}x : R \to (\pi, 2\pi)$; $\sec^{-1}x : (-\infty, -1] \cup [1, \infty) \to [0, \pi] - \left\{\frac{\pi}{2}\right\}$. Match maximum value of expression in Column-1 and minimum value of expression in Column-2 to corresponding values given in Column-3.

Column–1	Column–2	Column–3
(I) $\sin^{-1}x + \cos^{-1}x$	(i) $\tan^{-1}x + \cot^{-1}x$	(P) 0
(II) $\sin^{-1}(-x) + \sin^{-1}x$	(ii) $\tan^{-1}x + \tan^{-1}(-x)$	(Q) $\frac{\pi}{2}$
(III) $\cot^{-1}x + \cot^{-1}(-x) - \frac{5\pi}{2}$	(iii) $\sec^{-1}\left(\frac{1}{x}\right) + \cos^{-1}(x)$	(R) 2π
(IV) $\text{cosec}^{-1}\left(\frac{1}{x}\right) + \sin^{-1}x - \pi$	(iv) $\sec^{-1}x + \text{cosec}^{-1}x$	(S) $\frac{5\pi}{2}$

47. Which of the following combination is **CORRECT?**
(A) (I) (i) (P)
(B) (I) (ii) (R)
(C) (I) (i) (S)
(D) (II) (i) (R)

48. Which of the following combination is **CORRECT?**
(A) (II) (ii) (R)
(B) (III) (iii) (S)
(C) (IV) (iv) (P)
(D) (I) (i) (Q)

49. Which of the following combination is **CORRECT?**
(A) (III) (iv) (Q)
(B) (IV) (ii) (P)
(C) (II) (iv) (P)
(D) (I) (ii) (Q)

SECTION – B
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

50. If length of smallest path that can be traced from (0, 0) to $\left(\sqrt{2}+1,-1\right)$ without traversing inside the curve $x^2 + y^2 - 2\sqrt{2}x + 1 = 0$ is of the form $\frac{a\pi}{16} - b$; a, b $\in$ I, then value of a + b is equal to.

51. Let $f(a,b) = \sqrt{\left(\sqrt{25-a^2}-b\right)^2 + \left(a - \frac{1}{3}\sqrt{144-16b^2}\right)^2}$. If maximum and minimum value of f(a, b) is c and d respectively $\forall$ a, b $\in$ R, then c – d equals.

52. If there are only two points on the curve $2x^2 - 2y^2 - 4x + 21 = 0$ such that tangents drawn to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (a, b $\in$ N) are perpendicular, then a + b equals.

53. Number of solutions of the equation $\text{cosec}^2 x - \sec^2 x - \tan^2 x - \cot^2 x - \sin^2 x - \cos^2 x + 3 = 0$ $\forall$ x $\in [-2\pi, 2\pi]$ is.

54. If maximum value of $\tan^{-1}x + 2\sin^{-1}x + 3\sec^{-1}x$ is $\frac{p\pi}{q}$ where p, q are coprime integers, then p – q equals.